How Much Welfare Inequality Is Really Opportunity Inequality?

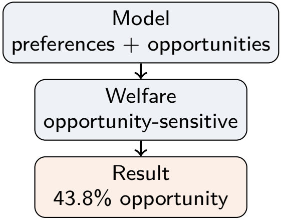
Structural Labor Supply, Latent Jobs, and Opportunity-Sensitive Welfare Decomposition

Hisham Haydar

Latent jobs & opportunities $\rightarrow$ opportunity-sensitive welfare $\rightarrow$ decomposition of welfare inequality

In one slide: model, welfare, decomposition, result

1. Estimate a structural labor-supply model with **preferences** + **latent opportunities**
2. Compute **opportunity-sensitive money-metric welfare**
3. Decompose welfare inequality into **opportunity** and **preference** components



Headline result:

Opportunities explain **0.039** Gini points, or **43.8%** of the explained component of welfare inequality

Outcomes do not reveal the same thing for everyone

- The same observed bundle can be a **chosen optimum** for one person and a **constrained second-best** for another
- If opportunities are omitted, standard models absorb them into **tastes**
- That distorts both the **positive** interpretation (preference heterogeneity) and the **normative** interpretation (who is truly worse off)
- This matters for how economists think about **compensation**, **responsibility**, and **redistribution**

Rich feasible set
$\Rightarrow$ *chosen optimum*

y

Truncated feasible set
$\Rightarrow$ *constrained*

same observed hours–income point

This paper sits at the intersection of four literatures

Latent jobs / RURO labor supply
Aaberge et al. (1995); Dagsvik et al. (2014); Dagsvik & Jia (2016); Capéau et al. (2016)

Money-metric welfare in discrete choice
Dagsvik & Karlström (2005); Bhattacharya (2015); Bargain et al. (2013)

This paper

Equality of opportunity / fairness
Roemer & Trannoy (2016); Ferreira & Gignoux (2011); Fleurbaey & Maniquet (2006)

Order-invariant decomposition
Shorrocks (2013)

The unusual feature is not structural labor supply alone, but treating the estimated opportunity mechanism as a first-class distributive primitive.

Existing papers stop too early; I connect behavior, welfare, and fairness

What they do	What I do
Constrained labor-supply papers stop at **fit, elasticities, or reform simulation**	Estimate **opportunities**, translate them into **welfare**, and **decompose welfare inequality**
Welfare papers with heterogeneous preferences stop before **explicit opportunity modeling**	Construct welfare **conditional on the estimated feasible set**, not on a universal menu
Equality-of-opportunity papers use **reduced-form circumstance partitions**	Use **structurally estimated job-opportunity distributions** as the circumstance object
Decomposition methods require **analyst-defined counterfactuals**	Define counterfactuals from the **estimated preference and opportunity blocks**

The contribution is integration: a latent-jobs labor-supply model, an opportunity-sensitive welfare metric, and a structural decomposition of welfare inequality in one design.

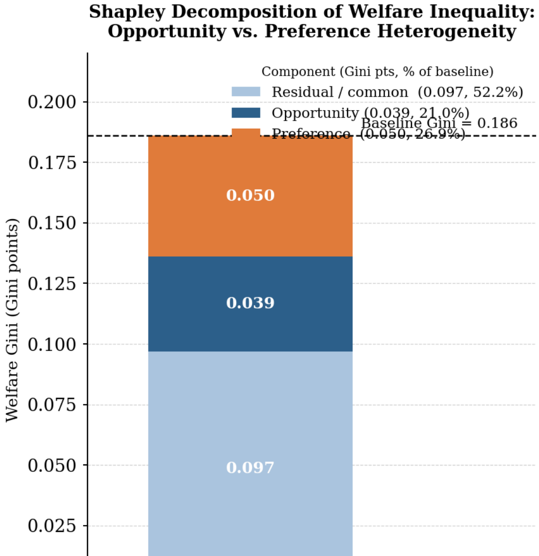
Belgium is the clean first application

- Data: **Belgian EU-SILC** + EUROMOD baseline tax-benefit rules
- Sample: **3,284** childless single women aged **25–54**
- 16 job packages: non-employment + 5 hours bins × 3 wage terciles
- Opportunity heterogeneity: region × education
- Preference heterogeneity: age, age^2, two latent taste types

Statistic	**Value**
Employment rate	77.6%
Mean weekly hours	34.9
Mean hourly wage	€15.8
Mean disp. income	€1,694
Flanders	56.8%
Wallonia	30.9%
Brussels	12.3%
Low education	25.9%
Medium education	42.5%
High education	31.6%

Design strips out joint household choice, childcare, and self-employment — keeping the first paper focused on the opportunity-vs-preference question.

Unequal opportunities explain a large share of welfare inequality



Scenario	Welfare Gini
Baseline	**0.186**
Opp. equalized	0.151
Pref. equalized	0.140
Both equalized	0.097

Component	Gini pts	% expl.
Opportunity	**0.039**	**43.8%**
Preference	**0.050**	56.2%

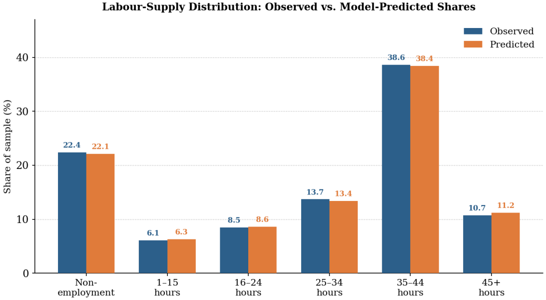
Opportunities: **21.0%** of baseline welfare Gini
Income Gini 0.164 vs welfare Gini **0.186**

The raw data already show structured opportunity differences

Employment rate by region × education

	Flan.	Wall.	Bxl.
Low ed.	72.4%	64.0%	**62.4%**
Medium ed.	79.3%	73.6%	71.2%
High ed.	**86.5%**	83.8%	82.4%

- Employment ranges from **62.4%** to **86.5%**
- Non-employment: **22.4%** of the sample
- Standard-hours work is the dominant mode



A credible model must fit these region × education gradients, not just overall participation.

The 16 job packages make latent opportunities operational

- Alternatives: non-employment + 5 hours bins $\times$ 3 wage terciles
- Hours bins: 1–15, 16–24, 25–34, 35–44, 45+
- The **preference block** ranks accessible packages
- The **opportunity block** governs access

Equalizing opportunities means equalizing access to a defined distribution of wage–hours packages, not shifting an abstract parameter.

	Low w.	Mid w.	High w.
1–15 h			
16–24 h			
25–34 h			
35–44 h			
45+ h			

+ non-employment

Pref. $\downarrow$ $\downarrow$ Opp.

Identification is simple to state: ranking versus access

- The identification challenge is to separate **how people rank packages** from **which packages are available**
- Region $\times$ education disciplines the opportunity block
- Age and latent taste types discipline the preference block
- Welfare is then computed **conditional on the feasible set**, not on a universal menu

Preference heterogeneity is meaningful, but not a catch-all residual

Parameter	**Est.**	**SE**
Log disp. income	2.946***	0.188
Log leisure, Type 1	1.428***	0.116
Type 2 shift	0.612***	0.141
Age/10 $\times$ log leisure	0.084***	0.029
$(\text{Age}/10)^2 \times$ log leisure	-0.010***	0.003
Work fixed cost	-1.324***	0.182
Type 2 logit intercept	-0.447***	0.118
Type 2 pop. share	0.390***	0.028

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

- Income enters positively; leisure matters strongly
- Two latent types: leisure coeff. **1.43** (T1); **2.04** (T2)
- Type 2 share: **39%**
- Work fixed cost is sizable (-1.324)
- Marginal value of leisure rises with age, concavely

Preferences do not vanish once opportunities are modeled. In the decomposition, preferences still explain slightly more than opportunities.

The opportunity estimates reveal a real labor-market structure

Log total offer intensity ($\ln q_i$)

Parameter	**Est.**	**SE**
Wallonia	−0.150**	0.061
Brussels	−0.192***	0.071
Low education	−0.108**	0.051
High education	0.151**	0.062
Wallonia × Low ed.	−0.109*	0.060
Brussels × Low ed.	−0.126*	0.071

Implied relative offer intensity
Ref: Flanders / Medium = 1.00

	Flan.	Wall.	Bxl.
Low	0.90	0.69	**0.65**
Medium	1.00	0.86	0.83
High	**1.16**	1.01	0.99

- Total offer intensity lowest for low-educated women outside Flanders
- Range: **0.65** to **1.16**
- Standard-hours packages much more prevalent than fringe-hours jobs
- Top-wage packages relatively scarcer

The opportunity block is not a nuisance term; it is an estimated, economically interpretable map of unequal access to work.

The benchmark misses exactly the gradients the opportunity block explains

Model comparison

	Full	**Bench.**
Pseudo-R^2	**0.271**	0.214
Exact hit rate	**39.2%**	33.8%
Empl. classif.	82.4%	—

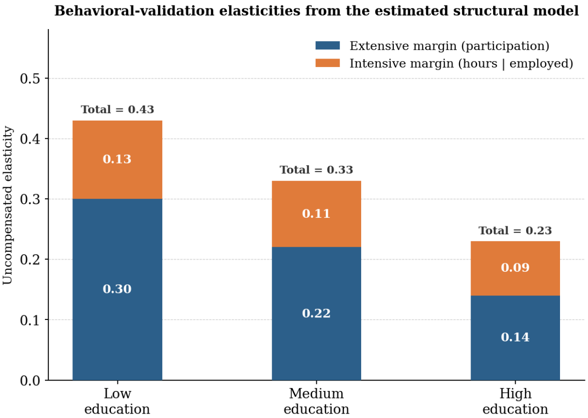
Employment rate: observed vs predicted

Cell	**Obs.**	**Pred.**
Flan./Low	72.4	73.0
Flan./Med	79.3	78.9
Flan./High	86.5	85.9
Wall./Low	64.0	64.8
Wall./Med	73.6	74.0
Wall./High	83.8	83.0
Bxl./Low	62.4	63.5
Bxl./Med	71.2	72.0
Bxl./High	82.4	81.7

- Full model improves pseudo-R^2 from 0.214 to **0.271**
- Exact hit rate: 33.8% $\rightarrow$ **39.2%**
- Main gain: recovering **region** $\times$ **education gradients** the benchmark compresses
- If the benchmark flattens those gradients, it absorbs them into **preferences**

Explicit opportunities improve fit precisely where the omitted-opportunity problem should matter most.

Elasticities validate the structure, but do not drive the contribution



Uncompensated arc elasticities from a 1% proportional wage increase, holding job-offer probabilities fixed

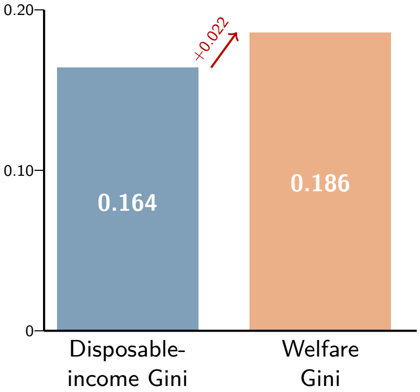
	Ext.	**Int.**	**Total**
Overall	0.21	0.11	**0.32**
Low ed.	0.30	0.13	**0.43**
Med. ed.	0.22	0.11	0.33
High ed.	0.14	0.09	0.23

This is a behavioral-validation result. The paper's contribution remains the welfare measure and its decomposition.

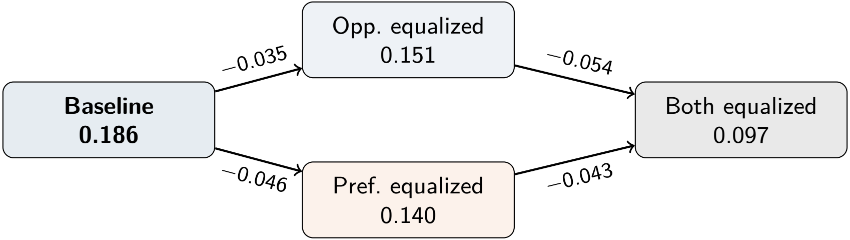
Welfare inequality is not the same object as income inequality

Welfare distribution	**Value**
Mean monthly welfare	€1,718
Median monthly welfare	€1,681
10th percentile	€1,079
90th percentile	€2,452

- Two individuals with similar income can have different welfare once one reaches that outcome from a much poorer feasible set
- Welfare inequality **rises** when we evaluate people not only by what they earn, but by what they can access



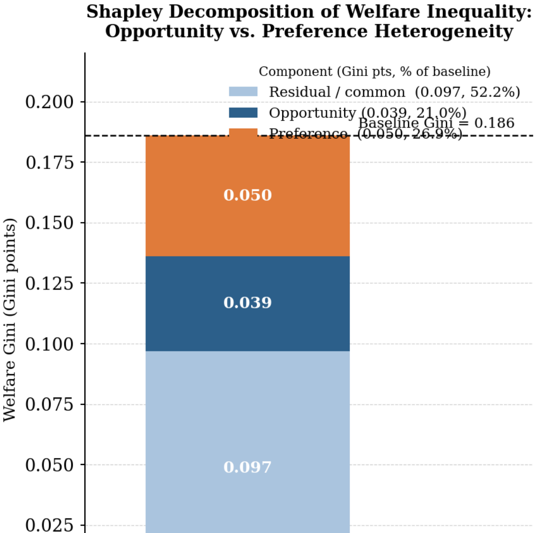
The decomposition is built from transparent counterfactuals



- Each counterfactual welfare Gini comes from an explicit equalization exercise
- The attribution uses an **order-invariant Shapley–Shorrocks** rule
- No dependence on the sequence in which factors are removed

The decomposition rests on explicit structural equalization exercises, not on a black-box regression residual.

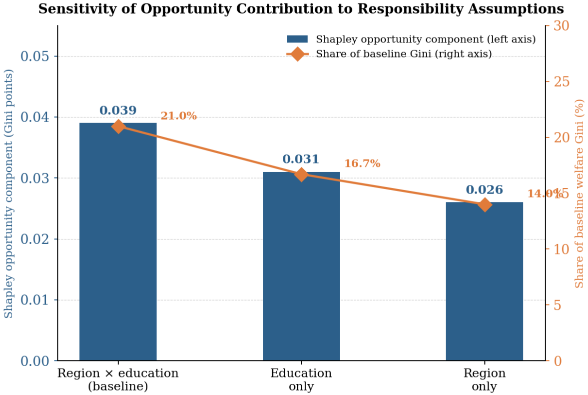
The main decomposition result is balanced, not rhetorical



Component	Gini pts	% explained
Opportunity	**0.039**	**43.8%**
Preference	**0.050**	**56.2%**
Residual/common	0.097	—
Baseline Gini	0.186	

- Opportunities: **21.0%** of baseline welfare Gini
- Preferences: **26.9%** of baseline welfare Gini
- The result is strongest *because* it is balanced
- Opportunities matter a great deal, but the paper does not claim they explain everything

The opportunity contribution is criterion-dependent, but does not disappear

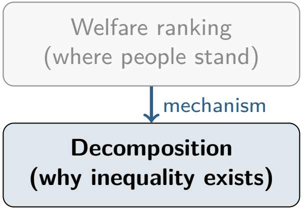


Compensable definition	Opp. comp.	% of Gini
Region × ed. (baseline)	**0.039**	**21.0%**
Education only	0.031	16.7%
Region only	0.026	14.0%

- The opportunity contribution is **criterion-dependent**
- It declines as the responsibility stance tightens
- But it does not vanish: **14–21%** of baseline Gini
- The contribution is the **decomposition itself**, not one preferred fairness metric

Decomposition matters more than welfare levels alone

- A welfare level tells us **where** people stand
- A decomposition tells us **why** inequality exists
- Policy implications differ if inequality is driven by **job access** rather than **preferences**
- That is why the paper's main object is a **decomposition**, not a ranking



Papers like Bargain et al. (2013) show welfare rankings are sensitive to preferences and metric choice. This paper delivers a mechanism question instead.

The Belgium baseline is a paper; Europe is an agenda

This paper
Belgium baseline
one coherent
decomposition

$\xrightarrow{\text{portability}}$

Next stage
Europe via EUROMOD
multi-country
scale-up

- Belgium provides the clean **first proof of concept**
- EUROMOD portability makes multi-country extension **feasible**
- Scaling too early risks a **ranking exercise**
- Europe belongs in the broader agenda, not in the paper's identity

The paper is narrow by design, not narrow by ambition

What I narrow	Why that strengthens the paper
Opportunity block is parsimonious (region $\times$ education)	Keeps identification transparent; avoids overfitting
Sample is clean (childless single women)	Removes joint household choice, childcare, self-employment
Decomposition uses two factors only	Shapley attribution is exact and interpretable
One country baseline	Establishes the mechanism before scaling up

Strategy: interpretability first, scale-up second

What the committee should remember

Question
How much welfare inequality is attributable to unequal job opportunities once labor supply is modeled as choice among latent jobs?

Method
A completed structural labor-supply paper with a latent-jobs opportunity block, opportunity-sensitive welfare, and Shapley decomposition

Result
Opportunities explain **0.039** Gini points (**43.8%** of explained welfare inequality); robust to responsibility assumptions (**14–21%** of baseline Gini)

Broader agenda
Scale the same pipeline beyond Belgium via EUROMOD; richer opportunity definitions; broader populations

Appendix / Backup Slides

Appendix 1: Full Named-Paper Literature Map

Strand	Named anchors
Latent jobs / RURO labor supply	Aaberge, Colombino & Strøm (1995); Dagsvik, Jia & Orsini (2014); Dagsvik & Jia (2016); Capéau, Decoster, De Sadeleer, Orsini, Pesslé & Tojerow (2016)
Money-metric welfare in discrete choice	Dagsvik & Karlström (2005); Bhattacharya (2015); Bargain, Decoster, Dolls, Neumann, Peichl & Siegloch (2013)
Equality of opportunity / fairness	Roemer & Trannoy (2016); Ferreira & Gignoux (2011); Fleurbaey & Maniquet (2006); Fleurbaey & Schokkaert (2009)
Order-invariant decomposition	Shorrocks (2013); Audoly, Fortin & Lemieux (2025)

The paper's location: estimate latent opportunities → compute opportunity-sensitive welfare → Shapley–Shorrocks decompose welfare inequality + responsibility sensitivity.

Appendix 2: What They Do / What I Do (Expanded)

Strand	**What they do**	**What I do**
Constrained labor supply	Estimate job-package models; use for fit, elasticities, reform simulations	Use the same structure as *input* into a welfare distribution; decompose welfare inequality via counterfactual opportunity equalization
Money-metric welfare	Develop/apply EV/CV welfare objects; welfare rankings across groups/countries	Construct welfare *conditional on heterogeneous feasible sets*, so constrained bundles are not misread as low-welfare choices
Equality of opportunity	Provide ethical language; measure between-type inequality with reduced-form partitions	Operationalize opportunities as *estimated job-opportunity distributions*; deliver responsibility-sensitive decomposition
Decomposition	Provide Shapley-value attributions once counterfactuals are defined	Define counterfactuals in *structural* terms and implement a two-factor Shapley–Shorrocks attribution of the welfare Gini

Appendix 3: Sample Construction

Step	Remaining
Belgian EU-SILC baseline population	—
Restrict to women aged 25–54	—
Drop partnered / married	—
Drop women with dependent children	—
Drop self-employed	—
Final estimation sample	3,284

Sample statistic	Value
Mean age	39.7
Employment rate	77.6%
Mean weekly hours (empl.)	34.9
Mean gross hourly wage	€15.8
Mean monthly disp. income	€1,694

Design restrictions remove avoidable sources of complexity while preserving sufficient heterogeneity for the opportunity block.

Appendix 4: Full Structural Estimates

Preference block

Parameter	Est.	SE	
Log disp. income	2.946	(0.188)	***
Log leisure, T1	1.428	(0.116)	***
T2 shift	0.612	(0.141)	***
Age/10 × log leis.	0.084	(0.029)	***
$(\text{Age}/10)^2$ × log leis.	−0.010	(0.003)	***
Work fixed cost	−1.324	(0.182)	***
T2 logit intercept	−0.447	(0.118)	***
T2 pop. share	0.390	(0.028)	***

Opportunity block — offer intensity

Parameter	Est.	SE	
Wallonia	−0.150	(0.061)	**
Brussels	−0.192	(0.071)	***
Low education	−0.108	(0.051)	**
High education	0.151	(0.062)	**
Wall.×Low	−0.109	(0.060)	*
Wall.×High	0.011	(0.071)	
Bxl.×Low	−0.126	(0.071)	*
Bxl.×High	0.031	(0.079)	

Package composition Ref: 1–15 h; low wage

16–24 h	0.214	(0.053)	***
25–34 h	0.548	(0.061)	***
35–44 h	1.012	(0.069)	***
45+ h	0.392	(0.079)	***
Mid wage	0.094	(0.041)	**
Top wage	−0.181	(0.051)	***

Appendix 5: Benchmark Comparison by Cell

Employment rate: observed vs predicted

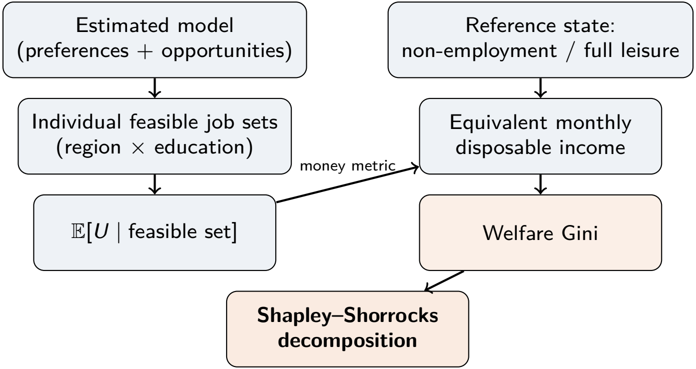
Cell	Obs.	Pred.
Flan./Low	72.4%	73.0%
Flan./Med	79.3%	78.9%
Flan./High	86.5%	85.9%
Wall./Low	64.0%	64.8%
Wall./Med	73.6%	74.0%
Wall./High	83.8%	83.0%
Bxl./Low	62.4%	63.5%
Bxl./Med	71.2%	72.0%
Bxl./High	82.4%	81.7%

Goodness-of-fit comparison

	Full model	Benchmark
Log-likelihood	−4,786.2	—
Pseudo-R^2	0.271	0.214
Exact hit rate	39.2%	33.8%
Empl. classif.	82.4%	—
Hours-bin hit rate	64.8%	—

The full model recovers the strong region × education gradients that the benchmark compresses.

Appendix 6: Welfare Construction in One Picture



Welfare ≡ the monthly disposable income at the reference state that makes the individual indifferent to her estimated constrained-choice environment. Comparable across individuals while respecting heterogeneous opportunity sets.

Appendix 7: Full Decomposition Table

Counterfactual welfare Gini

Scenario	**Gini**	**Δ vs baseline**
Baseline	0.186	—
Opp. equalized	0.151	−0.035
Pref. equalized	0.140	−0.046
Both equalized	0.097	−0.089

Shapley–Shorrocks attribution

	Abs.	**% expl.**	**% Gini**
Opportunity	0.039	43.8%	21.0%
Preference	0.050	56.2%	26.9%
Residual/common	0.097	—	52.2%

Welfare distribution summary

Mean	Median	P10	P90	Inc. Gini	Welfare Gini
€1,718	€1,681	€1,079	€2,452	0.164	0.186

Appendix 8: Responsibility-Sensitivity Table

Compensable definition	**Opp.-eq. Gini**	**Both-eq. Gini**	**Shapley opp. comp.**	**Share of expl. red.**	**Share of baseline**
Region $\times$ ed. (baseline)	0.151	0.097	0.039	43.8%	21.0%
Education only	0.158	0.105	0.031	38.9%	16.7%
Region only	0.164	0.110	0.026	34.2%	14.0%

- Tightening the responsibility stance reduces the compensable opportunity share, as expected
- Even under the strictest stance, unequal opportunities still explain about **14%** of baseline welfare inequality
- The contribution is the decomposition under *explicit* responsibility assumptions, not one preferred criterion

Appendix 9: Elasticity Table and Definition

Definition. Uncompensated arc elasticities from a 1% proportional increase in the gross wage attached to all working packages, holding the estimated opportunity mechanism (job-offer probabilities) fixed.

Group	Extensive margin	Intensive margin	Total hours
Overall sample	0.21	0.11	0.32
Low education	0.30	0.13	0.43
Medium education	0.22	0.11	0.33
High education	0.14	0.09	0.23

- Most heterogeneity appears on the **extensive margin**
- Lower-education women are more participation-responsive, consistent with weaker opportunity access
- A 10% wage increase would move employment from about 77.6% to 79.2% and mean hours from 34.9 to 35.3
- This is a **behavioral-validation** object, not the paper's headline contribution